Experimental evidence for negativity without negation[1]

Lisa HOFMANN — *University of Stuttgart*

Abstract. Approaches to negativity tags, like English *neither*-tags, *'not even'*-tags, and *why not*, differ in whether they treat the antecedent as a clause with clausal negation (e.g., Klima, 1964; Kramer and Rawlins, 2009; Brasoveanu et al., 2013) or, more broadly, one introducing a salient counterfactual proposition into the discourse (Hofmann, 2022, 2023a). This paper presents experimental evidence for the latter view, where negativity tags were rated as natural (compared to an affirmative baseline) when their antecedent was introduced without clausal negation, but in an anti-veridical embedding (e.g., *It's a lie*), under neg-raising (e.g., *I don't get the feeling*), or with 'lexical indicators of irony' (Horn, 2016, e.g., *as if*). The naturalness ratings also show fine-grained differences across possible licensors, suggesting additional pragmatic constraints and raising questions about the salience of the antecedent propositions.

Keywords: negation, anaphora, discourse negativity.

1. Introduction

This paper investigates a well-known but theoretically underexplored case of anaphoric polarity-sensitivity: the licensing of anaphoric *negativity-tags* (Klima, 1964). These are complex anaphoric expressions that require their antecedent to be *negative* in some sense. While these expressions are licensed following negative utterances (1a), they are infelicitous following affirmative ones (1b), even when the two are truth-conditionally equivalent. They include (2a) English *neither*-tags, and (b) *not-even*-tags. The same contrast has been observed for (c) factive uses of elliptical *why-not*-questions[2] (Hofmann, 2018, 2022, 2023a; Anand et al., 2021).

(1) a. Mary did**n't** fail the test.
b. Mary passed the test.

(2) a. Yes, and **neither** did Sue.
b. Yes, **not even** after getting really sick.
c. Yes, and it's pretty clear **why not**.

These tags depend on propositional content in discourse: *neither*-tags have an additive presupposition (Stockwell et al., 1968; see also Heim, 1992; Rullmann, 2003; Ahn, 2015; Thomas, 2022; Herburger and Portner, 2024 on *either*); *not-even*-tags have a scalar presupposition (Rooth, 1985; Wilkinson, 1996; Schwarz, 2005; Collins, 2016); and *'why not'* involves clausal ellipsis (Kramer and Rawlins, 2009; Hofmann, 2018, 2022; Anand et al., 2021), which has been analyzed as involving contextual entailment of elided content (Merchant, 2001; Kroll, 2020).

Standard semantic treatments of propositions as sets of possible worlds do not encode the polarity of the clause that introduced them. Accordingly, the antecedent utterances in (1a) and (1b) characterize the same set of worlds. The contrast is therefore unexpected under a purely semantic view on which negativity-tags target only unstructured propositional content, raising the question of what makes an utterance able to license an anaphoric negativity-tag.

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[2]Factive uses of *why not* appear in information-seeking questions or veridical contexts. They differ from the modal reading, illustrated below, found in (zero-answer) rhetorical questions.

(i) A: Let's go to the movies! B: Why not? (There is no reason that we wouldn't/shouldn't go to the movies.)

The dominant response to the licensing question characterizes the conditions for negativity-tags in terms of *sentential negativity*. On this view, an utterance counts as negative for anaphoric purposes if its clausal representation contains sentential negation, either syntactically or semantically (e.g., Klima, 1964; Ginzburg and Sag, 2000; Kramer and Rawlins, 2009; Farkas and Bruce, 2010; Brasoveanu et al., 2012, 2013, 2014; Roelofsen and Farkas, 2015). This approach originates in Klima's observation that negativity-tags are licensed by a range of negative expressions that vary in negative strength (in the sense of Zwarts, 1996) and syntactic position. Specifically, anti-additive quantifiers (*never, no one*), certain downward-entailing quantifiers (*rarely, few people*), and negative proximatives (*hardly, hardly anybody*) in adverbial positions (3a) or argument positions (3b–c) can license subsequent negativity-tags (4).

(3) Negative quantifiers:
  a. Pat {never/rarely/hardly} dances.
  b. {No one/few people/hardly anybody} dance(s).
  c. Sue dislikes {no one/few people/hardly} anybody.

(4) a. Neither did I.
  b. Not even a little bit.
  c. but I don't know why not.

The negative quantifiers in (3) have received a unifying analysis as contributing sentential negation and a positive (right upward-monotone) quantifier into a representation of their clause (e.g., Klima, 1964; Jacobs, 1980; Haegeman and Zanuttini, 1991; van den Berg, 1996; de Swart, 2000; Penka and Von Stechow, 2001; Horn, 2002; Zeijlstra, 2004).

(5) a. Pat never dances: $\neg$[Pat dances sometimes]
  b. Pat rarely dances: $\neg$[Pat dances often]

(6) a. No one dances: $\neg$[someone dances]
  b. Few people dance: $\neg$[many people dance]

On this basis, many accounts assume that negativity-tags are sensitive to syntactic or semantic reflexes of sentential negation (Klima, 1964; Ginzburg and Sag, 2000; Sanford et al., 2007; Kramer and Rawlins, 2009; Farkas and Bruce, 2010; Cooper and Ginzburg, 2012; Holmberg, 2013; Roelofsen and Farkas, 2015; Hofmann, 2018, 2023b). This central assumption of much previous theoretical work can be formulated as follows:

(7) **Sentential Negativity Hypothesis:** Negativity-tags are acceptable with antecedents whose clausal representation contains sentential negation.

This hypothesis underlies the widespread use of negativity-tags as a diagnostic for sentential versus constituent negation (e.g., McCawley, 1998; Huddleston and Pullum, 2002), and it is empirically successful for mono-clausal antecedents.The sentential view is challenged by cases where negativity-tags are licensed by embedded clauses without negation (Hofmann, 2022, 2023a). This includes embedded affirmative clauses under neg-raising (8a), including neg-raising in island contexts (8b), under anti-veridical[3] predicates (8c), or with ironic *as if* (8d).

[3](Giannakidou, 1998: 106) defines veridicality for a (unary) propositional operator O as follows: O is *veridical* iff,

(8) a. Neg-raising: I don't think that Sue danced at the party.
b. Island neg-raising: I don't get the impression that Sue danced at the party.
c. Anti-veridical predicates: It's a lie that Sue danced at the party.
d. Ironic *as if*: Sue is very shy. As if she danced at the party.

(9) a. ... but she didn't say why not.
b. ... not even a little bit.
c. ... and neither did Mary.

Although the antecedent clauses in (8) contain no clausal negation, the proposition that Sue danced is understood to be false. With anti-veridical predicates, this counterfactual inference is entailed; with neg-raising, it arises pragmatically (e.g., Bartsch, 1973; Gajewski, 2005; Romoli, 2013);[4] and with *like* and *as if*, it follows from the ironic interpretation (Giora, 1995; Horn, 2016).[5] Based on this generalization, Hofmann (2022, 2023a) more broadly characterizes discourse-negative utterances as ones introducing *counterfactual propositional content* – content that the speaker takes to be false, building on Krifka's (2013) proposal that polarity particles (PolPs, e.g. *yes/no*) can have propositional antecedents introduced by the prejacent of negation. This hypothesis can be stated as (10).

(10) **Discourse Negativity Hypothesis:** Negativity-tags are acceptable with antecedents which introduce propositional content into the discourse that is interpreted as counterfactual (based on literal content or pragmatic inference).

Both hypotheses predict that negativity-tags are licensed by antecedents with clausal negation and rejected after affirmative ones. Their predictions diverge for embedded antecedents that lack negation but introduce counterfactual content, as in (8). Here, the discourse-negativity hypothesis predicts licensing, whereas the sentential-negativity hypothesis predicts unacceptability.[6]

This paper reports on an experiment designed to address the following main research question:

(Q1) **Discourse vs. sentential negativity:** Are negativity-tags rated as natural following utterances that introduce counterfactual propositions without clausal negation?

The results are interpreted as supporting the discourse-negativity hypothesis: Negativity-tags are rated as natural, relative to an affirmative baseline, following counterfactual antecedents that lack clausal negation. Beyond this main research question, the experimental findings bear on two additional issues. First, previous work reports that counterfactual antecedents vary in how strongly they license negativity-tags (Hofmann, 2022). Second, negativity-tags differ in

for any proposition p: $O(p) \rightarrow p$ is logically valid; otherwise O is non-veridical; O is *anti-veridical* iff, for any proposition p: $O(p) \rightarrow \neg p$ is logically valid.

[4] Although syntactic analyses of classical neg-raising have recently been revived (Collins and Postal, 2014), these apply only to classical neg-raising predicates like *think* and *believe*. A pragmatic derivation is uncontroversial for neg-raising in (8c) (Collins and Postal, 2018).

[5] Horn (2016) characterizes ironic *like* and *as if* as 'lexical indicators of irony', that can license NPIs, in contrast to unmarked verbal irony, which does not.

[6] For classical neg-raising with *think/believe*, the prediction under the sentential view depends on the analysis. If such cases involve a covert negative element in the embedded clause (Collins and Postal, 2014), they could count as instances of clausal negation.

their presuppositional and anaphoric requirements. These considerations motivate two further research questions, concerning variability across licensors and tags.

In what follows, Section 2 summarizes relevant previous findings and introduces the secondary research questions. Section 3 presents an experiment designed to investigate whether English *neither*-tags, *not even*-tags, and *'why not'* are judged natural when following various counterfactual antecedents. The findings suggest that they are, supporting the discourse-negativity hypothesis, though sometimes with reduced naturalness ratings, raising important questions about the sources of variability in the responses. Section 4 concludes.

2. Building on previous studies

The basic contrast that negativity-tags are available after antecedents with clausal negation, but not after affirmative ones, has been confirmed experimentally for several related constructions: English *neither*-tags (Sanford et al., 2007), agreeing uses of negative PolPs in English (Brasoveanu et al., 2012, 2013; Roelofsen and Farkas, 2015) and German (Meijer et al., 2015; Claus et al., 2017), English agreeing utterances involving VP-ellipsis (Brasoveanu et al., 2013), and English polar question-tags (Sanford et al., 2007; Brasoveanu et al., 2014). The forced-choice continuation experiments of Sanford et al., 2007 and Brasoveanu et al. (2012, 2013, 2014) also found variability among various types of mono-clausal antecedents they tested: the proportion of negativity-tag choices varied with the semantic strength and syntactic position of the licensing expression, or with the presence of additional quantifiers in the clause.

Extending this paradigm to embedded environments, Hofmann (2022) provided preliminary evidence for discourse-negativity. Participants chose *why not* over the neutral counterpart *why* more often with antecedents introduced under anti-veridical predicates, neg-raising, or island neg-raising compared to an affirmative baseline. However, the proportion of *why not* choices was lower in island neg-raising and anti-veridical conditions than with overt negation and classical neg-raising. This variability also raises a question for the discourse-negativity hypothesis: if negativity-tags are licensed by counterfactual propositions, why do some counterfactual environments give rise to a lower proportion of negativity-tag follow-ups?

To begin exploring this issue, we consider three concerns raised by this pattern: First, forced-choice proportions may reflect relative preferences rather than acceptability, and naturalness ratings may yield a different pattern. The present study therefore uses naturalness ratings to test how reliable the by-licensor differences are. Second, in neg-raising contexts, matrix negation might spuriously be associated with the embedded clause, producing illusory licensing effects akin to NPI-illusions (e.g. Drenhaus et al., 2005). To address this concern, the present study includes ironic *like/as if*, which supports a counterfactual interpretation pragmatically without overt negation. Third, the previous study grouped several anti-veridical, neg-raising, and island neg-raising environments into conditions. Testing each possible licensor separately (as in experimental designs in Tonhauser et al., 2018; Degen and Tonhauser, 2025, on projection), allows for a more fine-grained assessment of differences between counterfactual environments. The experiment presented here therefore addressed the following secondary research question:

(Q2) **Variable negativity:** Do naturalness ratings for negativity-tags differ between the different licensing expressions?

Finally, since negativity-tags differ in their anaphoric and presuppositional requirements, it is possible that they show different patterns of acceptability across the different possible licensors (see also McCawley, 1998). The experiment therefore also addressed the following question:

(Q3) **Tag variability:** Do naturalness ratings differ for different negativity-tags?

3. Experiment

The experiment was designed to investigate whether English *neither*-tags, *not-even*-tags, and *why-not*-questions are judged natural continuations (i) after antecedents that introduce a counterfactual proposition in discourse without clausal negation, as predicted by the discourse-negativity hypothesis, or (ii) only after antecedents containing sentential negation, as predicted by the sentential-negativity hypothesis. It compared 15 possible licensors, listed in (11). They included a positive and negative control, and 13 embedding environments that differed in how they support the counterfactual inference that the embedded clause is false.

(11) a. positive control: *I think that*
b. negative control: *I think that… not…*
c. 2 neg-raising contexts (*I don't think/believe that*)
d. 4 island neg-raising contexts
(*I don't get the feeling/impression that, It's not my view/opinion that*)
e. 5 anti-veridical environments
(*I disagree/heavily doubt that, It's* {*a lie / just a rumor*} *that, You're wrong that*)
f. 2 lexical indicators of irony (*Like…, As if…*)

3.1. Method

Design. The dependent variable was the naturalness RATING of negativity-tags following the possible licensors in (11). Participants responded on a slider labeled 'totally unnatural' and 'totally natural' at the ends (coded `0-1`). The design crossed two predictors: LICENSOR (15 levels), which encoded which of the bi-clausal antecedent types in (11) was presented; and TAG (3 levels), which encoded the negativity-tag being rated: *neither*, *not even*, or *why not*.

Materials and procedure. The experiment was run online on a self-hosted instance of jsPsych (de Leeuw et al., 2023; von der Malsburg, 2025). As shown in Figure 1, each trial presented a context sentence and a simple visual scene with two stick-figure characters.[7] A first button press revealed the licensor utterance in the left speech bubble; a second press revealed the negativity-tag response together with the rating prompt and slider.

Each of the 15 licensors was combined with one of 15 embedded clauses (items, listed in Appendix A), such as *the new daytime drama on TV grew its viewership last month* in Figure 1. Each item was paired with a matching context sentence and with versions of the three neg-tag continuations. For the irony conditions (*like*, *as if*), the licensor utterance included an additional sentence that made the literal reading implausible, as in (12), because previous studies found that this encourages an ironic interpretation (Jorgensen et al., 1984; Pexman and Olineck, 2002).

[7] Adapted from the webcomic *xkcd* (`https://xkcd.com`).

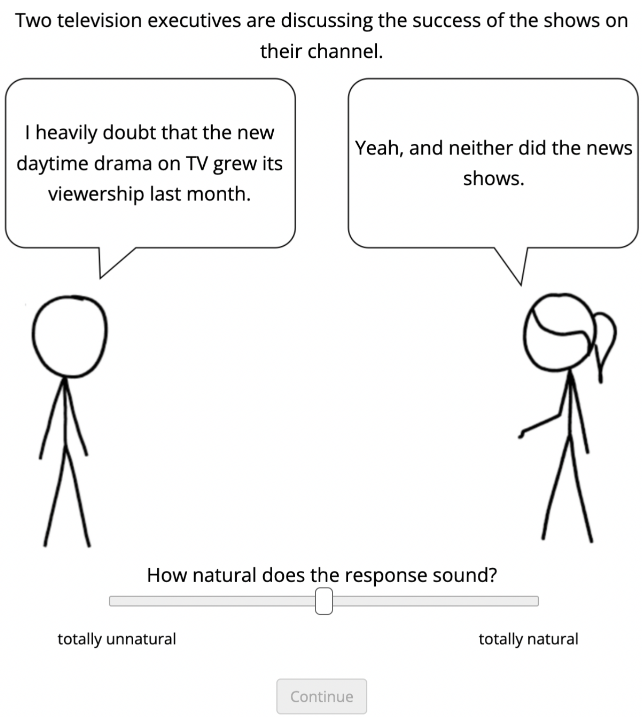

Figure 1: View for participants during naturalness rating

(12) Our new programming hasn't been successful. As if the new daytime drama on TV grew its viewership last month.

Each participant saw the 15 licensors in randomized order, each randomly paired with a unique item and with one of the tags, so that each tag appeared five times. At the end, participants filled out an optional demographic survey. To encourage truthful responses, participants were told that they would be paid no matter what they answered in the survey.

Participants and data exclusion. 266 participants were recruited on Prolific and compensated at a rate of $11.06GBP$ per hour (age range: 18–76, mean 42.57; gender: 115 f, 4 nb, 147 m). Median completion time was about 6:30 minutes. Data was excluded from seven participants who did not self-report as native speakers of American English, and from 49 participants who rated the positive control above 0.5. The remaining dataset contains 3195 observations from 213 participants (age range: 18–76, mean 43.35; gender: 99 f, 3 nb, 118 m).

3.2. Results and analysis

3.2.1. Discourse negativity vs. sentential negativity

We first evaluate the core question that distinguishes the two hypotheses: whether negativity-tags are judged natural following antecedents that introduce counterfactual propositional content without clausal negation. Figure 2 shows mean naturalness ratings by licensor, aggregated across tags, with 95% bootstrapped confidence intervals.

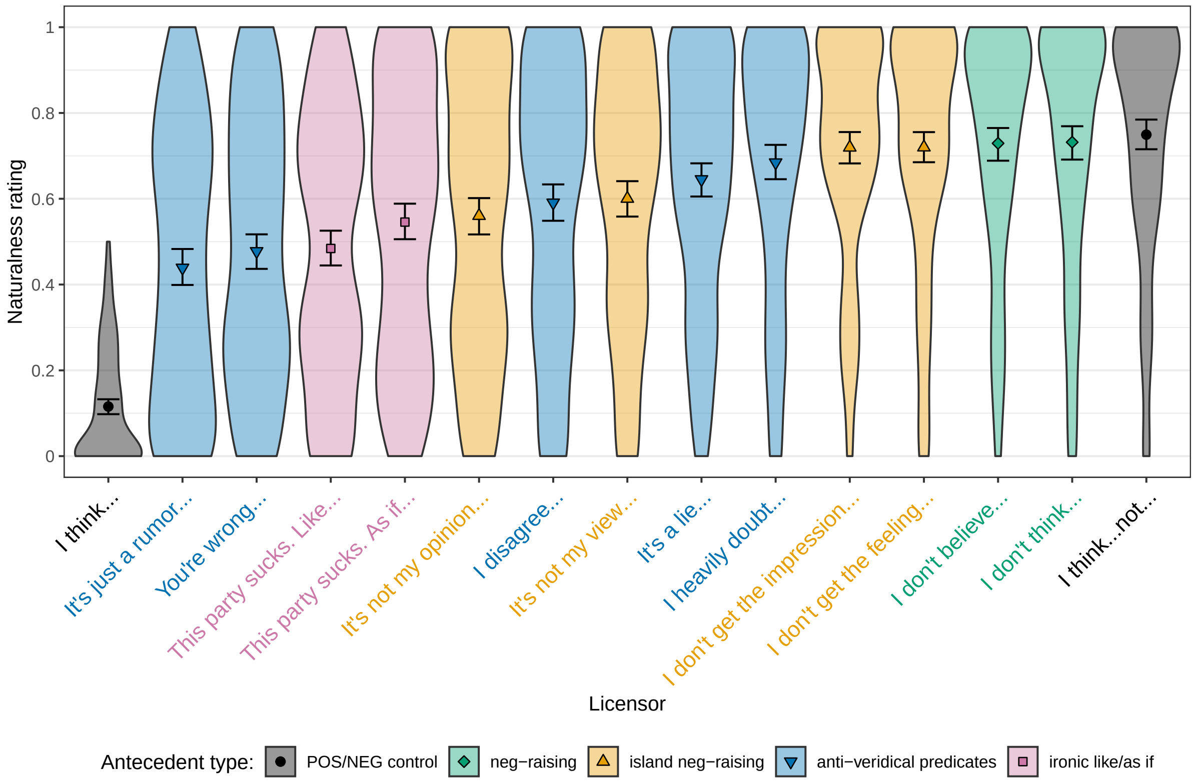

Figure 2: Mean naturalness ratings, and 95% bootstrapped confidence intervals by licensor. Violin plots indicate the kernel probability density of ratings in each condition.

The control conditions establish the expected baseline pattern: The positive control (*I think...*) yields low ratings, confirming that negativity-tags are generally infelicitous after affirmative antecedents. The negative control (*I think...not...*) yields the highest ratings, showing that overt sentential negation robustly licenses negativity-tags. The critical comparison concerns the embedding environments that lack clausal negation in the embedded clause but nevertheless introduce a counterfactual interpretation. As is visually apparent in Figure 2, all counterfactual licensors receive clearly higher ratings than the affirmative control. This includes neg-raising, island neg-raising, anti-veridical predicates, and ironic *like/as if*.

This difference was supported by a Bayesian mixed-effects beta regression analysis using `brms` (Bürkner, 2017) in `r` (R Core Team, 2025). The model predicted naturalness ratings[8] from fixed effects of LICENSOR (treatment-coded with the positive control as the reference level), TAG (sum-coded), and their interaction, and included random intercepts and slopes for participants and items. The main effect of LICENSOR showed that all counterfactual licensors had positive estimated effects relative to the positive control, with the 95% credible intervals of the corresponding coefficients excluding zero, indicating credibly higher ratings than in the affirmative baseline (e.g., *It's just a rumor*: $\mu = 1.52$, 95% CrI [1.26, 1.79]).[9]

[8]To model the ratings using a beta regression, ratings were transformed from the interval [0,1] to the interval (0,1), following Smithson and Verkuilen (2006), see also Degen and Tonhauser (2025).

[9]Additional model details are provided in Appendix B; the full output and analysis scripts are available in an OSF repository at `https://doi.org/10.17605/OSF.IO/EHSPV`.

3.2.2. Variable negativity

We next ask whether all counterfactual licensors license negativity-tags equally well. Figure 2 suggests that they do not: In particular, naturalness ratings are lower with anti-veridical predicates, ironic *like* and *as if*, and some island neg-raising environments, compared to the negative control. These differences were quantified in post-hoc pairwise comparisons of estimated marginal means for LICENSOR using the `emmeans` package (Lenth et al., 2024), based on the regression model described above. The pairwise differences are summarized in Figure 3.

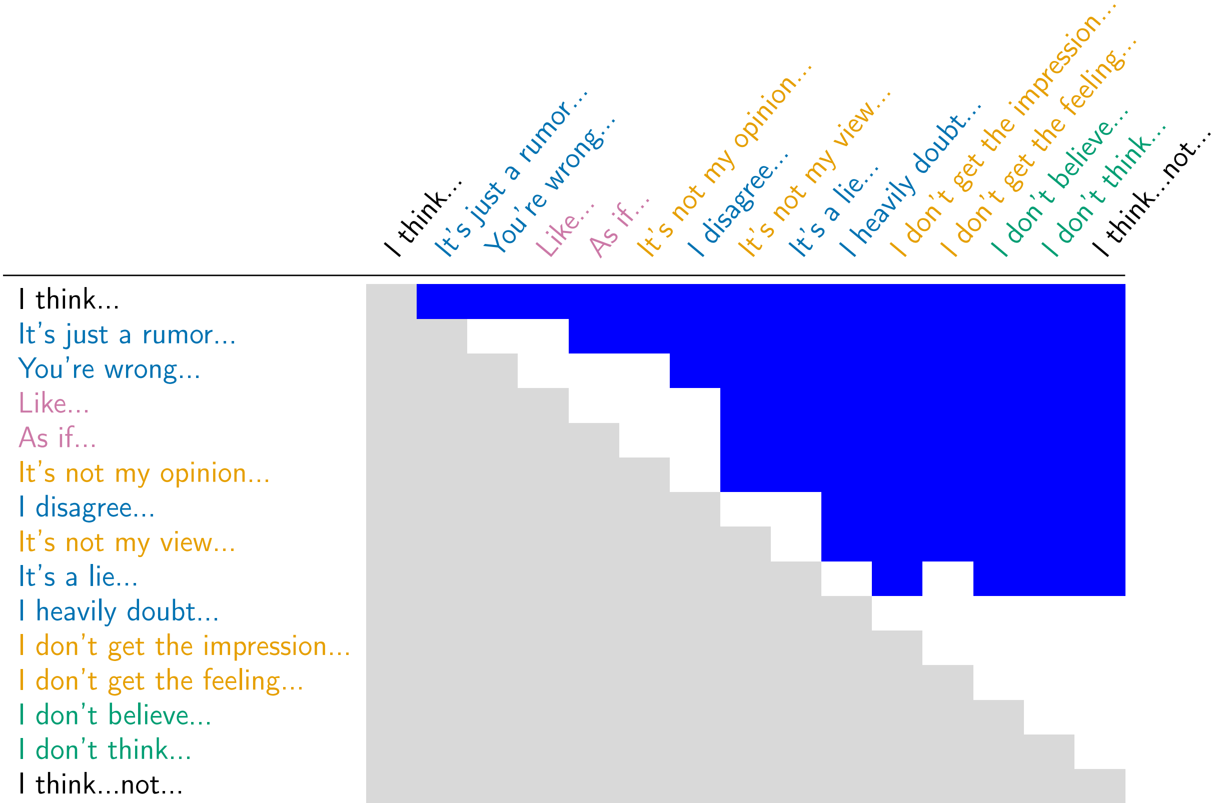

Figure 3: Pairwise differences between licensors, ordered from top to bottom and left to right by increasing mean naturalness rating. A white cell means that the 95% HDI of the row/column licensor pair includes 0, and a blue cell means that the 95% HDI does not include 0 and the row licensor received a lower rating than the column licensor.

Relative to the negative control, negativity-tags are rated credibly lower in the two irony conditions (*like*, *as if*), in two of the island neg-raising environments (*It's not my opinion/view*), and in four anti-veridical embeddings (*It's just a rumor...*, *You're wrong...*, *I disagree...*, *It's a lie...*). In contrast, classical neg-raising (*I don't think/believe...*) does not differ credibly from the negative control. The same holds for the island neg-raising cases *I don't get the feeling/impression...*, and for *I heavily doubt...*, which was grouped with the anti-veridical predicates.

In sum, these results indicate fine-grained differences across licensors, All counterfactual licensors increase the naturalness of negativity-tags relative to the affirmative baseline, but the strongest licensing is observed with clausal negation, classical neg-raising, island neg-raising with *I don't get the feeling/impression...*, and embeddings under *I heavily doubt...*. Anti-veridical predicates (aside from *heavily doubt*), lexical irony markers, and parts of the island neg-raising class yield lower ratings. This pattern suggests that a counterfactual propositional antecedent is necessary but not sufficient for maximal acceptability, and that additional semantic or pragmatic factors modulate how strongly an antecedent functions as discourse-negative.

3.2.3. Tag variability

Finally, we ask whether naturalness ratings differ between the three negativity-tags? Figure 4 plots mean ratings by tag, aggregated across licensors. Overall, the three tags are close in mean rating, and the regression model described above shows no main effect of TAG (sum-coded) relative to the grand mean. However, post-hoc `emmeans` comparisons (Figure 5) suggest a small but credible difference such that *why not* receives lower ratings than the other two tags.

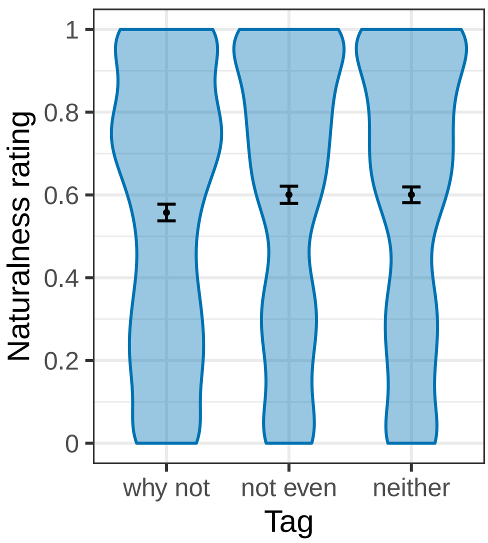

Figure 4: Means, 95% bootstrapped confidence intervals, and kernel probability densities for naturalness ratings by tag.

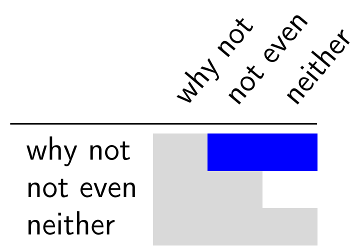

Figure 5: Pairwise differences between tags, ordered from top to bottom and left to right by increasing mean naturalness rating. A white cell means that the 95% HDI of the row/column tag pair includes 0, and a blue cell means that the 95% HDI does not include 0 and the row tag received a lower rating than the column tag.

At the same time, the tag-differences are not uniform across all licensors, and the overall disadvantage for *why not* appears to be driven by a small subset of conditions. In the regression model, all 95% credible intervals for the LICENSOR × TAG interaction coefficients included zero (with sum coding for TAG), not indicating a reliable overall interaction. However, post-hoc comparisons of the pairwise differences by LICENSOR and TAG (using `emmeans`) reveals a handful of licensor-specific contrasts (Figure 6). For example, *why not* is rated lower than *not even* with ironic *like*, *I heavily doubt…*, and with neg-raising under *believe*. It is also rated lower than *neither* with ironic *as if*, with *I heavily doubt…*, with island neg-raising under *It's not my view…*, and with neg-raising under *think*. For most licensors, though, no by-tag differences emerge. Those marginal differences that do emerge are quite small, as can be seen in Figure 7, below.

3.3. Discussion

Discourse vs. sentential negativity. The result that all counterfactual conditions yield higher naturalness ratings than the affirmative baseline supports the discourse-negativity hypothesis: If

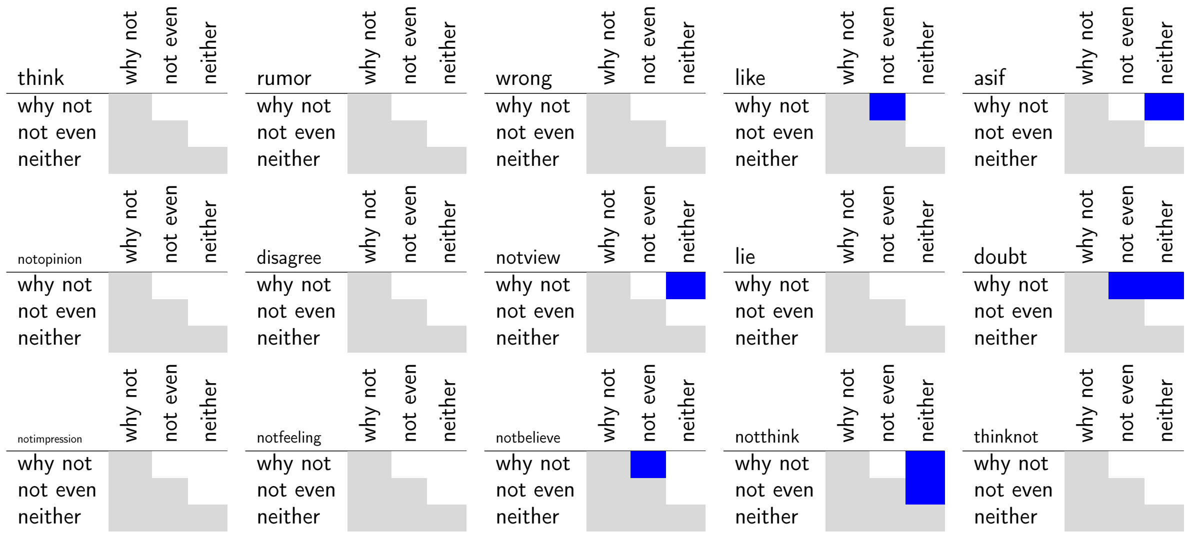

Figure 6: Pairwise marginal differences for between TAGS for each LICENSOR, ordered from left to right and top to bottom by increasing mean naturalness rating. A white cell means that the 95% HDI of the row/column tag pair for a given licensor includes 0, and a blue cell means that the 95% HDI does not include 0 and the row tag received a lower rating than the column tag.

negativity-tags were licensed only by clausal negation, antecedents under (island) neg-raising, anti-veridical predicates, and ironic *like/as if* should pattern with the positive control. Instead, all of these environments yield clearly higher naturalness ratings, indicating that negativity-tag licensing is sensitive to something present in these conditions – a counterfactual proposition.

The conditions where the counterfactual inference is derived pragmatically are particularly informative: First, the pragmatic derivation in (island) neg-raising and in the irony conditions, indicates that pragmatic reasoning can contribute materially to negativity-tag licensing. Second, the irony conditions introduce a counterfactual inference without any overt negation at all, ruling out concerns of spurious licensing mentioned in Section 2. Together, these points suggest an antecedent requirement defined over a discourse-level representation that can be enriched by pragmatic reasoning, beyond purely clausal representations.

Methodological implications for diagnosing clausal negation. The experimental findings suggest that the negativity-tags used here diagnose the presence of a counterfactual propositional embedding rather than clausal negation. This calls into question their widespread use as a diagnostic for clausal negation (e.g., McCawley, 1998; Pullum and Huddleston, 2002; Collins and Postal, 2017). Especially for multi-clausal antecedents, where various environments can introduce counterfactual inferences about embedded content in several ways, one cannot be sure that they target an antecedent that includes clausal negation. However, neg-tags can still serve as a diagnostic tool to distinguish sentential from constituent negation in mono-clausal antecedents, assuming that no other clause-internal anti-veridical propositional operators are present.

Variable negativity. The results reproduce and refine the earlier finding that some counterfactual environments yield weaker licensing (Hofmann, 2022). Negativity-tags are rated lower with several anti-veridical licensors, in the irony conditions, and in parts of the island neg-raising class, even though these contexts support a counterfactual interpretation.

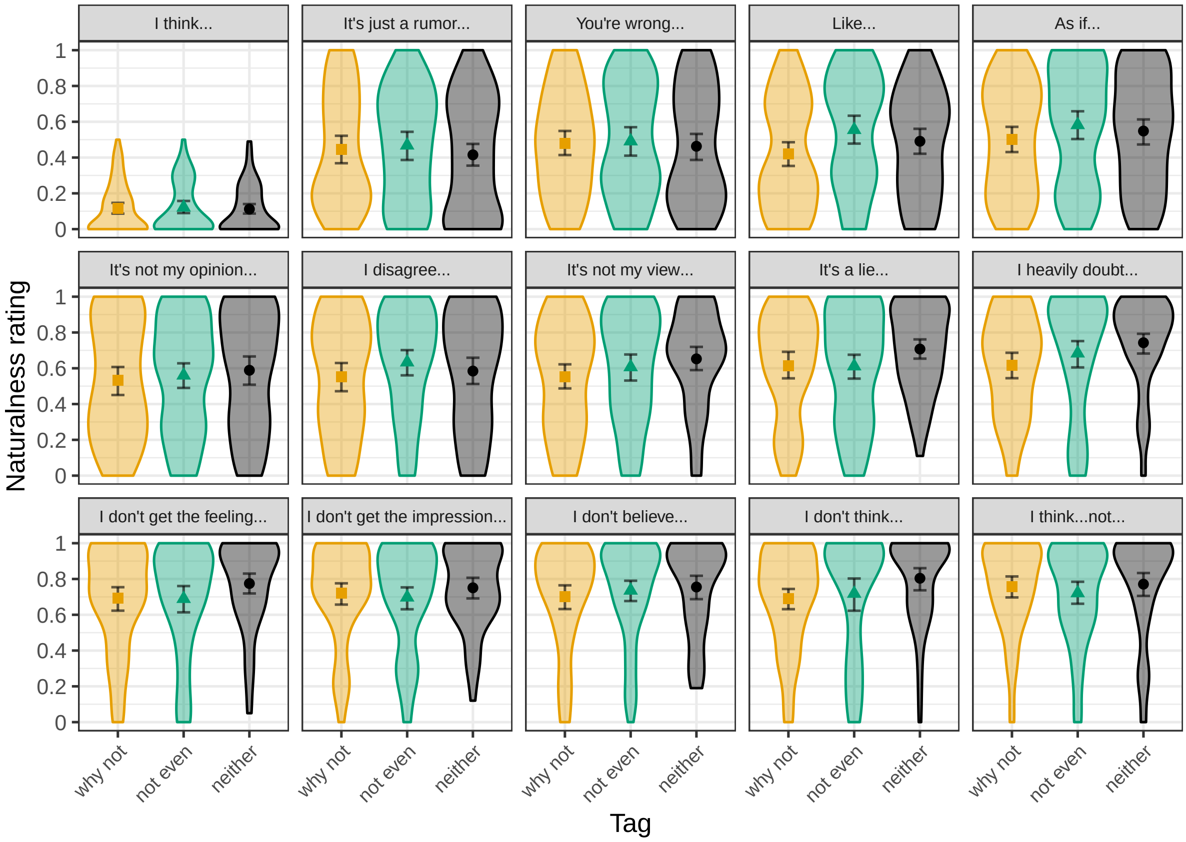

Figure 7: Means, 95% bootstrapped confidence intervals, and kernel probability densities for naturalness ratings by tag for each level of licensor.

The reduced ratings in these non-negated counterfactual contexts raise a challenge for the discourse-negativity hypothesis: if counterfactual content licenses negativity-tags, why is licensing by overt negation often stronger? One way to maintain the discourse-negativity view is to treat a counterfactual propositional antecedent as necessary for licensing, but not sufficient for maximal acceptability. Although the present study does not resolve this issue, it refines the empirical picture by examining each embedding environment separately rather than grouping them. Similar fine-grained variability is well documented in work on projection, where various factors (such as at-issueness, prior beliefs, and lexical meaning) modulate projection ratings (Tonhauser et al., 2018; Degen and Tonhauser, 2025). The variability observed here might similarly reflect pragmatic modulation of the availability of counterfactual propositions for negativity-tags.

A possible explanation could be that embedding environments differ in the prominence or accessibility of the counterfactual proposition for anaphora, which might be due to differences in at-issueness. While some authors suggest that anaphoric availability is largely independent of at-issueness (e.g., AnderBois et al., 2015; Snider, 2017), other works suggest links between polarity phenomena involving propositional anaphora and at-issueness (Horn, 2002; Krifka, 2013; Claus et al., 2017; Kroll, 2020). This raises the theoretical question of how propositional prominence may be incorporated, and possibly operationalized in terms of QUD-based discourse structure. A natural follow-up experiment would be to test whether reduced negativity in certain counterfactual environments correlates with lower ratings on at-issueness measures, as in work on projection (e.g., Tonhauser et al., 2018; Degen and Tonhauser, 2025).

Tag variability. Across tags, overall differences are small, but *why not* shows a small disadvantage, which appears to be driven by a subset of licensors. This suggests that tag-specific requirements can interact with particular licensors, even if differences are small.

4. Conclusion

Overall, the results support the discourse-negativity hypothesis. Negativity-tags are rated as natural, relative to an affirmative baseline, following antecedents that introduce counterfactual content without clausal negation. This suggests that negativity-tags are licensed by the counterfactual proposition that is present in these conditions, and that their antecedent requirement operates on representations that allow for pragmatic enrichment, beyond clausal representations.

An adequate account of discourse negativity must therefore integrate three components: It must allow for negativity without overt negation (e.g., anti-veridical predicates), for pragmatically derived counterfactuality (e.g., neg-raising, irony), and for standard sentential negation. Such a level of representation is provided in the dynamic semantic analysis of discourse-negativity and the negativity-tag *why not* in Hofmann (2022) and Hofmann (2023a). The observed variability across licensors, together with the limited tag-specific effects, further indicate that pragmatic factors modulate the strength of discourse-negativity as sufficient conditions on maximal naturalness. The question about the nature of these soft constraints on the kind of propositional anaphoricity present in negativity-tags is a key question for future work.

A. Experimental materials

(13) 15 embedded-clause items

- (i) the party planning committee made an effort this year.
- (ii) the volunteers performed many administrative tasks this month.
- (iii) the delegates narrowed down possible target dates in the final resolution.
- (iv) the warehouse crew received benefits in this period.
- (v) the students in my class improved their grades this semester.
- (vi) the defense lawyer's closing statement made an impact in this case.
- (vii) the park rangers fed the lions this morning.
- (viii) the pianists played Beethoven sonatas at the competition.
- (ix) the new daytime drama on TV grew its viewership last month.
- (x) the reporters interviewed the whistleblowers this morning.
- (xi) the home team warmed up their muscles before the game.
- (xii) the coal miners talked to the press this week.
- (xiii) the public universities had to follow the new tax laws.
- (xiv) the guests at our yearly banquet received champagne as an aperitif.
- (xv) the pharmacists discussed potential side effects of the medicine at the conference.

B. Statistical model

The data was analyzed using a Bayesian mixed-effects regression model. Because the dependent variable RATING consists of slider responses, we fit a beta-regression model. To meet the assumptions of the beta distribution, ratings were transformed from the closed unit interval [0,1] to the open interval (0,1) following the procedure proposed by Smithson and Verkuilen (2006); see also Degen and Tonhauser (2025). The model predicted transformed ratings for the following:

- Fixed effects: LICENSOR (treatment-coded, with the positive control as reference level), TAG (sum-coded), and their interaction.
- Random effects: correlated intercepts and by-TAG slopes for PARTICIPANT and correlated intercepts and by-LICENSOR slopes for ITEM
- Beta-dispersion (phi-parameter): Main effect of LICENSOR, and random intercepts for PARTICIPANT

Table 1 provides the estimated mean, standard deviations and 95% credible intervals (CRIs) for the posterior distributions of selected fixed effects (the main effects). The full model output can be found at `https://doi.org/10.17605/OSF.IO/EHSPV`.

	mean	std.dev	95% CRI lower	95% CRI upper
Intercept	-1.63	0.13	-1.89	-1.38
LICENSOR: **rumor**	1.52	0.13	1.26	1.79
LICENSOR: **wrong**	1.58	0.13	1.33	1.84
LICENSOR: **like**	1.66	0.18	1.31	2.01
LICENSOR: **as if**	1.77	0.13	1.51	2.03
LICENSOR: **not opinion**	1.80	0.13	1.54	2.06
LICENSOR: **disagree**	1.97	0.14	1.70	2.23
LICENSOR: **not view**	2.04	0.13	1.78	2.30
LICENSOR: **lie**	2.21	0.14	1.95	2.49
LICENSOR: **doubt**	2.40	0.14	2.13	2.66
LICENSOR: **not impression**	2.49	0.13	2.24	2.75
LICENSOR: **not feeling**	2.45	0.14	2.18	2.71
LICENSOR: **not believe**	2.53	0.15	2.24	2.82
LICENSOR: **not think**	2.50	0.14	2.23	2.77
LICENSOR: **think not**	2.51	0.14	2.24	2.78
TAG: **why not**	-0.01	0.12	-0.24	0.21
TAG: **not even**	0.07	0.13	-0.18	0.32

Table 1: Posterior means, standard deviations and 95% CRIs for selected fixed effects in the estimated mixed-effects logistic regression model.

The model was estimated using the `brms` package (Bürkner, 2017) in R (R Core Team, 2025), with low-information (default) priors. The MCMC estimation used 6 chains, 10000 iterations per chain, 2000 burn-in, and no thinning.

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